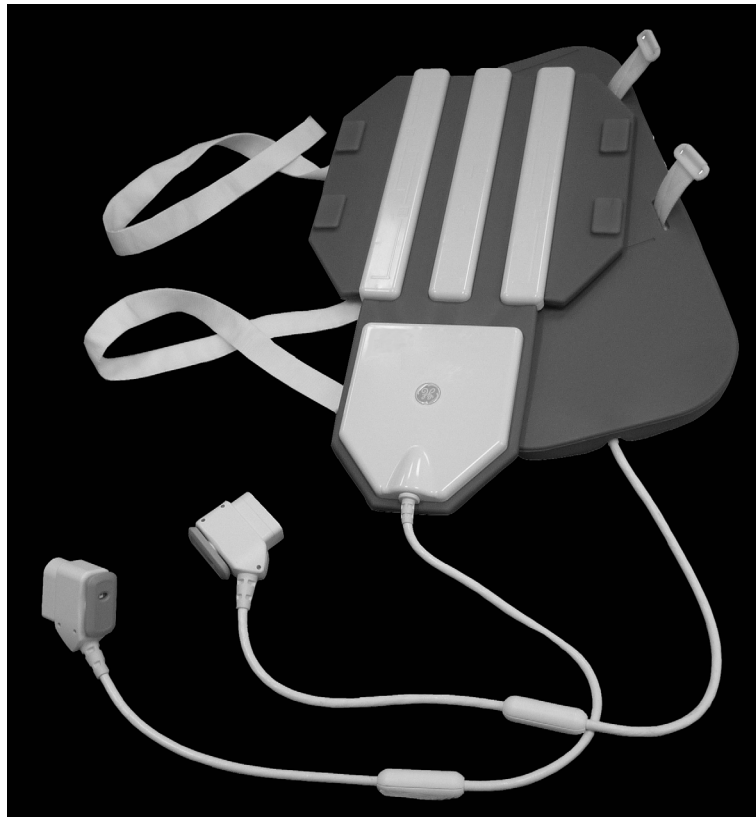


GE Healthcare

32-Channel Torso Array

Instructions for use



Distributed by GE Healthcare



NeoCoil
Pewaukee, WI, USA

32-Channel Torso Array
Instructions for Use. English
NC004IFU-EN
Rev 4 (08/09)

© 2009 GE Healthcare
All rights reserved.

This page intentionally left blank.

Medical device directive

This product conforms with the requirements of council directive 93/42/EEC concerning medical devices when it bears the following CE Mark of Conformity:



European Representatives:



NeoCoil
EMERGO EUROPE
Molenstraat 15
2513 BH, The Hague
The Netherlands
Telephone: +31.70.345.8570
Fax: +31.70.346.7299



CAUTION:

United States federal law restricts this device to sale by, or on the order of, a physician.



CAUTION:

This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.



WARNING:

This product contains chemicals, including lead, known to the state of California to cause birth defects or other reproductive harm. Wash hands after handling.

This page intentionally left blank.

Table of contents

Chapter 1: Introduction and coil specifications

Introduction.....	1-1
Shipping contents	1-2
Use of symbols.....	1-4
Coil classification	1-6
Coil description	1-6
Indications for use.....	1-7
Contraindications.....	1-7
Environmental requirements	1-7
Coil specifications.....	1-7
Coil installation and service.....	1-8
Product lifetime	1-8

Chapter 2: Safety

Patient safety	2-1
Emergency procedures	2-2
Equipment safety	2-2
Electrical and mechanical safety	2-3
Cautions.....	2-4
Precautions	2-4
Modifications	2-4
Reporting incidents.....	2-5
MR safety information.....	2-5

Chapter 3: Coil and patient setup

Positioning the patient and coil	3-1
Localizing anatomy	3-3
Recommended scan protocols.....	3-3
Recommended transportation method.....	3-4

Chapter 4: Quality assurance

Scanner verification	4-1
Coil imaging performance verification.....	4-1
Coil and phantom setup.....	4-1
Multi-coil quality assurance (MCQA) tool.	4-3

Chapter 5: Coil maintenance

Introduction.....	5-1
Coil care	5-1
Accessories.....	5-2
Contact information.....	5-3

This page intentionally left blank.

Chapter 1

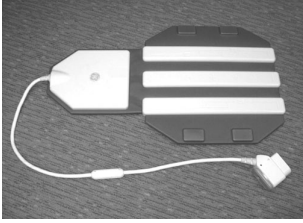
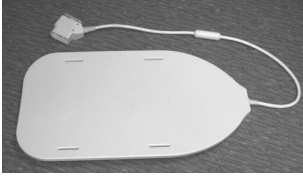
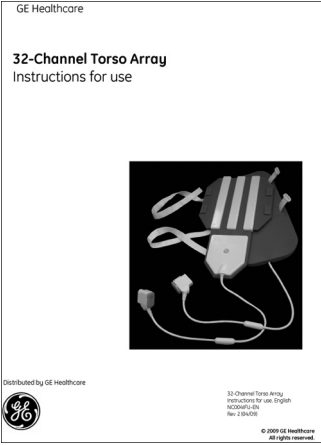
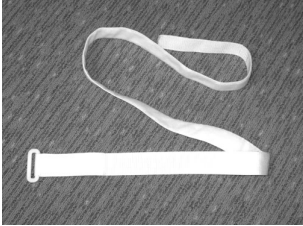
Introduction and coil specifications

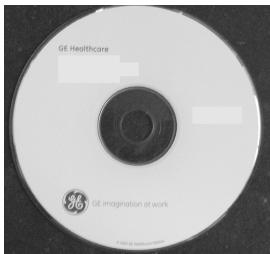
Introduction

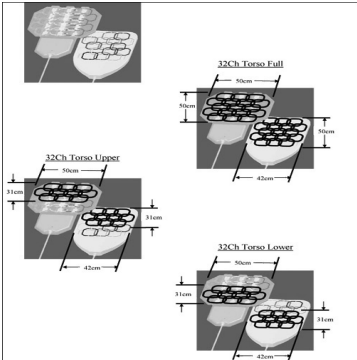
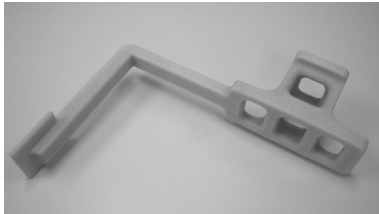
This manual describes the safety precautions, features, use and care of the 3.0T 32-Channel Torso Array, compatible with GE 3.0T MRI scanners. Please review this manual thoroughly before using the device for imaging. Refer to your MR scanner manual in conjunction with this manual

If you have questions or comments on this manual, or need assistance with the use of this product, please contact your GE Healthcare representative at: 1-800-437-1171 or go to our website at www.gehealthcare.com.

Shipping contents

Part name	NeoCoil part #	GE part #	Qty.
3.0T 32-Channel Torso Array - Anterior 	NC004001	5339708-2	1
3.0T 32-Channel Torso Array - Posterior 	NC004010	5339708-3	1
Instructions for Use 	NC004IFU	N/A	1
Strap assembly 	NC004034	N/A	2
Posterior patient pad 	NC004033	N/A	1

Part name	NeoCoil part #	GE part #	Qty.
Chin pad 	NC004006	N/A	1
Anterior patient pad 	NC004007	N/A	1
Cable pad 	NC004008	N/A	1
Coil definition disk 	N/A	5339708-7	1
Service manual 	NC004SVM	N/A	1

Part name	NeoCoil part #	GE part #	Qty.
Quick reference guide 	NC004089	N/A	1
Phantom positioner 	NC004009	5339708-5	1

Use of symbols

Coil users should understand the following symbols used for caution and warning explanations:

Symbol	Description
 CAUTION:	Indicates a hazard that may cause a minor injury or equipment damage.
 WARNING:	Indicates a hazard that may cause serious injury.
	Do not cross or loop coil cables.

Symbol	Description
	Dispose of RF coil properly.
	Manufacturer.
	Manufactured date.
	Consult Instructions for Use.
	Temperature limitation.
	Humidity limitation.
	Atmospheric pressure limitation.
	CE Mark of Conformity

Coil classification

Symbol	Classification
	Class II equipment
	Type BF applied
	IPXO Ingress Protection rating
	Suitable for continuous operation
	Non-sterile device

Coil description

The 3.0T 32-Channel Torso Array is a receive-only coil designed to give optimum signal-to-noise ratio and uniform coverage of the torso. The anterior coil is padded with a soft material designed to maximize patient comfort.

The coil design enables the clinical user to collect high resolution images of the chest, abdomen, and pelvic regions.

The coil uses several modes of operation (Table 1-1).

Table 1-1 Modes of operation

Modes of operation	S/I	# of channels	Coils used
32Ch Torso Full	Ant 50cm x 50cm, Post 42cm x 50cm	32	Anterior and Posterior
32Ch Torso Upper	Ant 50cm x 31cm, Post 42cm x 31cm	20	Anterior and Posterior
32Ch Torso Lower	Ant 50cm x 31cm, Post 42cm x 31cm	18	Anterior and Posterior
32Ch Torso Pelv	Ant 37cm x 22cm, Post 30cm x 22cm	8	Anterior and Posterior
32Ch Torso Pos	Post 42cm x 50cm	16	Posterior
32Ch Torso Ant	Ant 50cm x 50cm	16	Anterior and Posterior
32Ch Torso UpAnt	Ant 50cm x 31cm	10	Anterior and Posterior

The coil description in the scanner interface is “32Torso Array by NeoCoil.” Proper patient positioning will produce the best results. Improper use of the coil is the major cause of image artifacts.

Indications for use

To be used in conjunction with GE 3.0T MRI scanners to produce diagnostic images of the torso that can be interpreted by a trained physician.

Contraindications

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR system (check the MR Safety Manual for additional contraindications):



WARNING:

Scanning is contraindicated for patients who have electrically, magnetically, or mechanically activated implants (e.g. cardiac pacemakers) because the magnetic and electromagnetic fields produced by the MR system may interfere with the operations of such a device.



WARNING:

Scanning patients with intracranial aneurysm clips or other ferromagnetic metal implants is contraindicated.

Environmental requirements

Store the coil flat in an air conditioned scan room or equipment room.

Table 1-2 Coil environmental requirements.

	Transport/storage	Operating conditions
Barometric pressure	86 kPa to 106 KPa	
Humidity	20% to 95% non-condensing	45% to 75% non-condensing
Temperature	-15° C to 55° C (5° F to 131° F)	21° C to 32° C (69.8° F to 89.6° F)

Coil specifications

Weight:

- Anterior: 10.3 lbs. (4.7 kg)
- Posterior: 15.5 lbs. (7.0 kg)

Dimensions (excluding system cables):

- Anterior: 36.2" L x 21.1" W x 1.5" H (91.9 cm x 53.6 cm x 3.8 cm)
- Posterior: 32.5" L x 17.3" W x 2.5" H (82.5 cm x 43.9 cm x 6.4 cm)

Field strength	Model	Part #	Frequency
3.0T	3.0T 32-Channel Torso Array	Anterior GE: 5339708-2 NeoCoil: NC004001 Posterior GE: 5339708-3 NeoCoil: NC004010	127.72 MHz

Coil installation and service

Coil installation and configuration must be done by your GE Service Representative. Please refer to the service manual for installation and service information.

Product lifetime

The product lifetime of the 3.0T 32-Channel Torso Array is three years from the date of shipment from the manufacturer.

Chapter 2

Safety

Patient safety

Patient safety and comfort must be your primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards with the potential to cause injuries.

**WARNING:**

Scanning is contraindicated for patients who have electrically, magnetically, or mechanically activated implants (e.g. cardiac pacemakers) because the magnetic and electromagnetic fields produced by the MR system may interfere with the operations of such a device.

**WARNING:**

Scanning patients with intracranial aneurysm clips or other ferromagnetic metal implants is contraindicated.

**WARNING:**

Patients with ferromagnetic metal implants should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.

**WARNING:**

Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the system's manufacturer.

**WARNING:**

There is a risk to scanning feverish or decompensated cardiac patients.

**WARNING:**

Prevent all cables from forming loops and coming into contact with the patients. Contact and looping may result in burns to the patient.





WARNING:

Do not allow the surface coil cable to touch the patient. Patient burns may result.



WARNING:

Use only GE equipment (e.g. gating equipment, vital signs monitoring systems and RF coils) that has been specifically tested and approved for use in the MR environment. Equipment that has not been approved may result in burns or other injuries to the patient. Such equipment may also interfere with proper operation of this coil.

Route cables through the center of the magnet bore. Place cables under the cushion whenever possible. Routing near the sides of the bore increases the likelihood of cable heating from induced currents.

Keep the length of the cable in the bore to a minimum. Avoid bending the cable 180 degrees. Route cables out of the bore in the most direct way possible, without looping, coiling, or crossing of cables.

Use a thermal resistant material or pad to keep the cable from touching the patients.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. If the patient reports heating, burning or tingling sensations, **STOP THE SCAN IMMEDIATELY.**

Emergency procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress
- Remove the patient from the scan room if medical treatment is needed.

Equipment safety

All personnel using this coil must be instructed in its proper connection and handling. Personnel must observe all warnings and cautions appearing in this Manual.



WARNING:

This coil is not suitable for use in the presence of flammable anesthetics or in combination with air, oxygen or nitrous oxide.



CAUTION:

Do not lift or carry the coil by the cable. Damage to the coil and/or cable could result.

Electrical and mechanical safety

**WARNING:**

This coil consists of electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Only properly trained and qualified personnel should service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.

Before using the coil, visually check it to ensure that there is no external damage. Do not use the coil if the housing or cable is broken.

**CAUTION:**

Prevent cables from looping. Looping may cause RF coupling and degrade the coil's scan performance.

**CAUTION:**

Coil contains electro-static discharge (ESD) sensitive components. Avoid contact with connector's contact pins. ESD may degrade scan performance of the coil.

Cautions

The following general statements apply to scanning with a magnetic resonance system:



CAUTION:

- Remove unplugged surface coils or accessory devices from the magnet bore. They may cause patient burns.
- Remove blue wire-type ECG gating cables from the magnet bore before scanning with surface coils.
- Do not use blue wire-type ECG leads when performing gated surface coil scans. Use peripheral gating, high impedance EDG leads, or thin film ECG leads rather than blue wire-type ECG leads when performing surface coil scans.
- Make sure the patient does not touch the magnet bore while scanning.
- Do not use the coil with exposed metal parts or damaged insulation.
- Do not allow the cables to touch the patient. Use thermal resistant material or a pad to keep the cable from touching the patient.
- Cables should not be looped or crossed. Arcing and patient burns could result.
- Route all cables out of the magnet bore so that they do not contact the patient.
- Facial make-up should be removed before scanning because eye make-up may contain metal flakes that can cause skin and eye irritation. Permanent eye-liner tattoos may cause eye irritation due to the presence of ferromagnetic particles.
- Patients who work in environments with a risk of having embedded metallic fragments in or near the eye should be carefully screened before having an MR exam.
- All precautions, cautions, instructions and recommendations provided by the manufacturer of the MRI system will apply.

Precautions

Precautions should be taken when scanning patients with the following conditions:

- Greater than normal potential for cardiac arrest,
- An increased likelihood for developing seizures or claustrophobia,
- Unconscious, heavily sedated, confused patients or those with whom no reliable communications can be maintained.

NOTE: For further details, review the warnings included in your MR system's safety manual.

Modifications

Any changes or modifications to this coil must be approved and performed by your GE Healthcare Service representative.

Reporting incidents

Users should contact GE Healthcare immediately to report an incident and/or injury to a patient, operator or maintenance employee that occurred as a result of coil operation.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

MR safety information

Please refer to the MR Safety Guide, GE direction #2381696. The MR Safety Guide describes the safety information you and the physicians must understand thoroughly before you begin to use the MR system. If you need additional training, seek assistance from qualified GE Healthcare personnel.

This equipment is intended for use by qualified personnel only.

This manual should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this manual before attempting to use this product.**

This page intentionally left blank.

Chapter 3

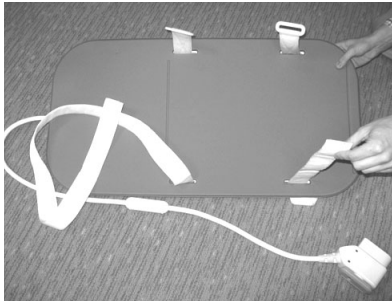
Coil and patient setup

Positioning the patient and coil


WARNING:

Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from RF coupling of this device with extraneous devices in the magnet.

Table 3-1 Positioning the patient and coil on the scan table.

Visual	Description
	1. Place the patient pad on the posterior coil and thread the straps through the holes at one edge of the pad and posterior coil. Guide the straps under the posterior coil and up through the holes at the opposite edge of the posterior coil and pad.
	2. Position the posterior coil component with patient pad and straps on the patient table. Ensure that the straps hang loosely over the side of the table. 3. Place additional pads along the remaining table length, resulting in a flat surface.

Visual	Description
	4. Position the patient on the coil, feet-first, supine, with the region of interest lying on the posterior section of the coil. Ensure that the cable is situated on the end opposite the patient's head. 5. If using the posterior coil only, position coil and patient as above. Do not position the anterior coil. Secure the straps snugly around the patient.
 	6. Position the anterior section of the coil over the anatomical region of interest. For patient comfort or to better position the coil over the region of interest, patient may be positioned with arms up or at the sides. Align the anterior and posterior sections by the alignment marks in the housings. Skip if using only the posterior coil. 7. Fasten the straps for a snug and comfortable fit. Place chin pad on the anterior coil. Make sure comfort pads are in place. 8. Provide earplugs to the patient after all instructions have been given.

**WARNING:**

Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99dB(A). Hearing protection must have a Noise Reduction Rating (NPR) of 28dB or better.

Visual	Description
	9. When using both posterior and anterior coils, plug them into only P1 and P2, or only P3 and P4 (any other combinations are unacceptable configurations). If using only the posterior coil, plug it into any receptacle. The coil cables should always be routed directly to the coil port and should not form loops.
	10. Coils for the GE Scanners with P-port connectors require the type of connector shown at left.

CAUTION:

Depending on how the coil and patient are situated on the table, there may not be enough slack in the cable to allow the table to be lowered when the coil is plugged in. Attempting to lower the table with insufficient slack in the cable may cause coil damage.

Localizing anatomy

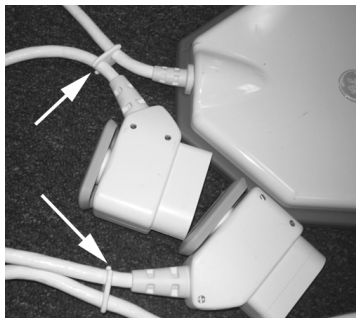
Visual	Description
	Landmark on the coil elements corresponding to the anatomical region of interest.

Recommended scan protocols

Use the system recommended protocols for scanning with this coil for optimum results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

Recommended transportation method

Visual	Description
 A close-up photograph showing the ends of white cables being secured with white plastic cable clips. Two white arrows point to the clips, indicating where they should be applied to prevent the cables from dragging on the ground.	For transport and storage, the ends of the cables should be secured with the cable clips (arrows) to prevent cables from dragging on the ground. The cables should not be clipped while scanning.
 A photograph showing a person's hands using white lifting straps to lift a large, flat, rectangular coil assembly. The straps are routed through the holes of the posterior coil, demonstrating the correct method for transporting the equipment.	The straps can be used to carry the coils away from the table. Ensure that the straps are securely fastened and routed through the holes of the posterior coil prior to lifting.

Chapter 4

Quality assurance

Scanner verification

Perform system level Signal to Noise Check. Refer to Service Methods CD>Functional Checks>RF Path>Signal to Noise Check

Coil imaging performance verification

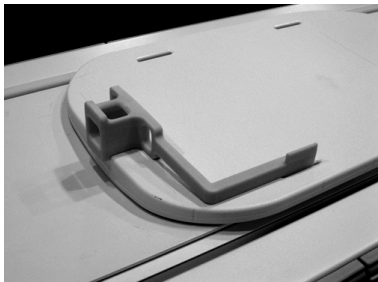
Table 4-1 Tools required:

Description	Part Number	Qty.
Phantom positioner	5339708-5	1
Phantom	5341595 / 2381682-10 / 2406200 / 5325953-6	2

Coil and phantom setup

Record the serial number of the coil(s) being used. The coil system cable must be directed towards the magnet and be plugged into port P1 or P2 for this test. The MCQA tool will be run separately for the posterior and anterior coils.

Table 4-2 Coil and phantom setup - 1

Visual	Description
	1. Remove any other surface coils (if present) from the cradle
	2. Place the posterior coil and phantom positioner on the table as shown. Remove any pads present on the coil. The cable must be directed towards the magnet and be plugged into port P1 or P2.

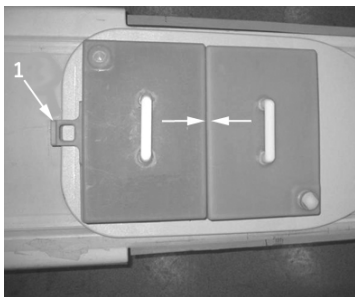
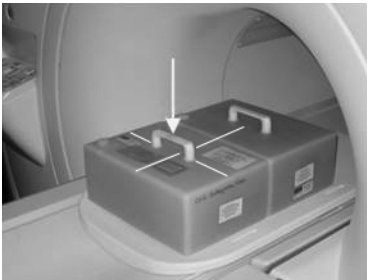
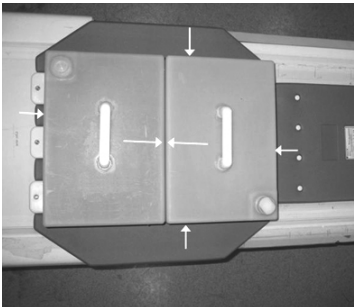
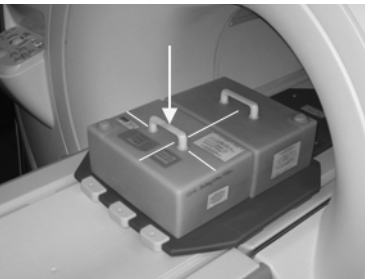
Visual	Description
	3. Using the phantom positioner (1), center the phantoms in the R/L and S/I directions. Ensure that the phantoms are aligned with each other and that they are touching each other.
	4. Remove the phantom positioner. Landmark on the center of the phantom handle furthest from the coil port as shown.

Table 4-3 Coil and phantom setup - 2

Visual	Description
	1. Remove any other surface coils (if present) from the cradle
	2. Place the anterior coil and phantoms on the table as shown. The cable must be directed towards the magnet and be plugged into port P1 or P2. Ensure that the phantoms are centered in the R/L direction and also centered between the white plastic caps. The phantom positioner is not needed.
	3. Landmark on the center of the phantom handle furthest from the coil port as shown.

Multi-coil quality assurance (MCQA) tool.

CAUTION: All RF coil related tests must be run on a well-calibrated system that has passed all system tests, particularly white pixel, correlated noise, and MCR (multi-coil receive) tool.



WARNING: To prevent hearing impairment, hearing protection is required for all people in the scan room during a scan. Acoustic levels may exceed 99 dB(A). Hearing protection must have a Noise Reduction Rating (NRR) of 28 dB or better.



The MCQA tool will be run separately for the posterior and the anterior coils.

To run the MCQA tool for the posterior coil, setup the posterior coil according to the "Coil and phantom setup 1" Table 4-2

To initiate MCQA:

1. From the MR service desktop window, Image quality tab, select Multi-Coil QA Tool, as shown in Figure 4-1.

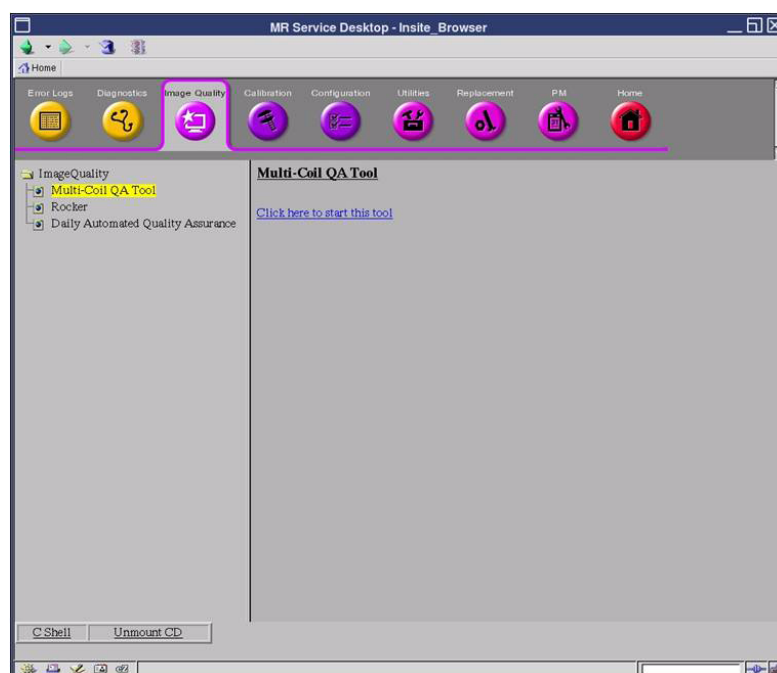


Figure 4-1 Select MCQA tool from the MR service desktop window

2. On the test side, select "Click here to start the tool"

3. A pop-up window as shown in Figure 4-2 will appear with the 'Current Coil' field automatically filled in. Please wait for a few seconds if the field is empty.

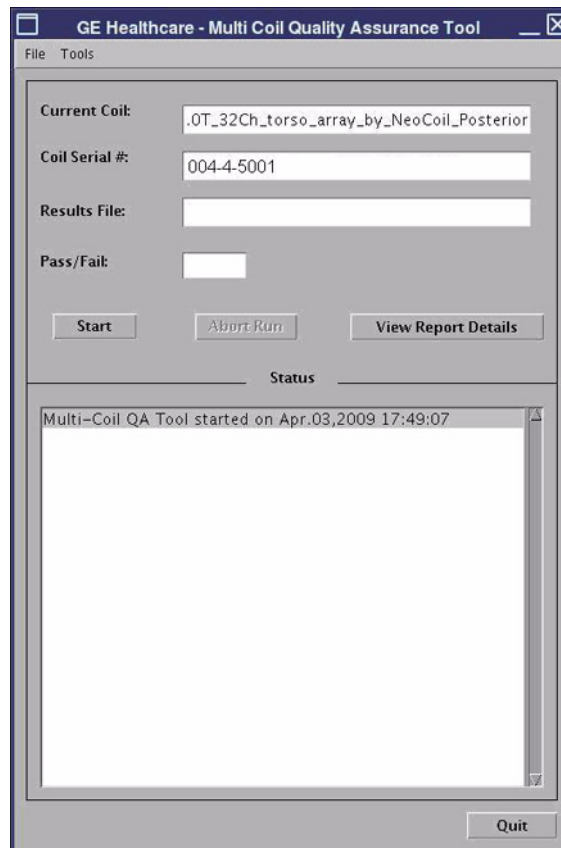


Figure 4-2 MCQA tool window with the current coil connected

4. Enter the coil serial number.
5. Click **[Start]** to initiate the automated test.
6. When the test is completed, results are displayed (see Figure 4-3). The results window shows the integrated Signal-to-Noise Ratio (SNR) as a function of the receive channel number. The PASS/FAIL status should show PASS if all coil elements are functioning properly.

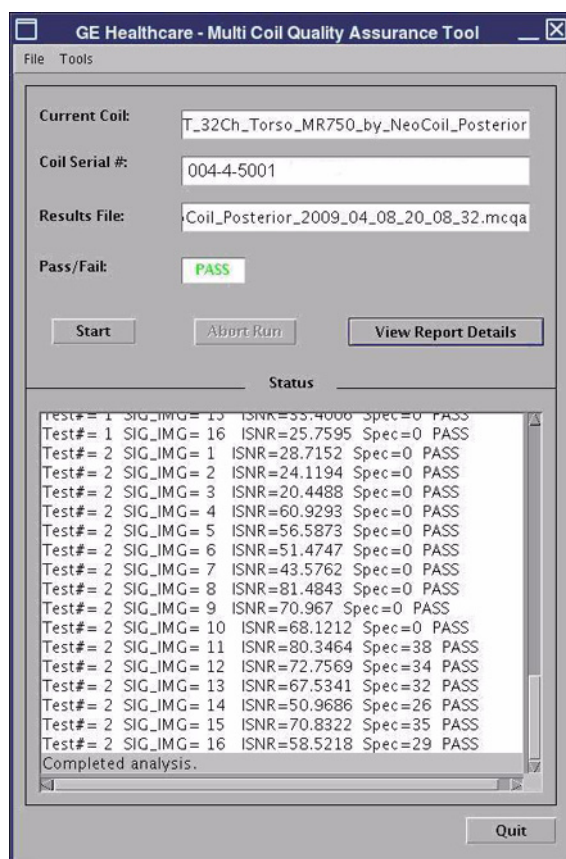


Figure 4-3 MCQA tool displaying the test results for the posterior coil

- Upon startup of the MCQA tool, or during execution of the test procedure, you may see the following phantom placement and landmarking message (see Figure 4-4). Click on **Yes** to continue the test, or **No** to cancel the test so that you can make any adjustments or corrections that may be required.



Figure 4-4 Click on **Yes** to continue the test, or **No** to cancel the test

To run the MCQA tool for the anterior coil, setup the anterior coil and phantom according to the "Coil and phantom setup 2" Table 4-3

To initiate MCQA:

- From the MR service desktop window, Image quality tab, select Multi-Coil QA Tool, as shown in Figure 4-1.
- On the test side, select "Click here to start the tool"
- A pop-up window as shown in Figure 4-5 will appear with the 'Current Coil' field automatically filled in. Please wait for a few seconds if the field is empty.

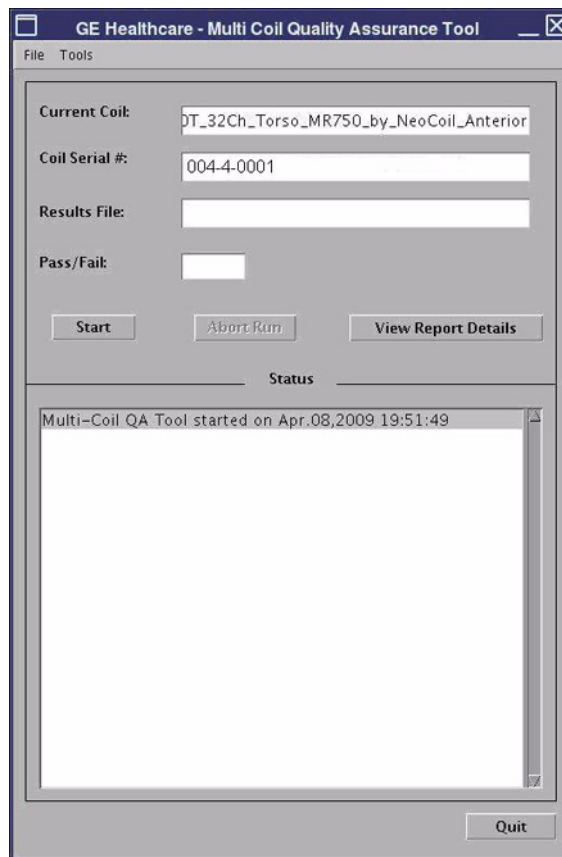


Figure 4-5 MCQA tool window with the anterior coil connected

4. Enter the coil serial number.
5. Click **[Start]** to initiate the automated test.
6. When the test is completed, results are displayed (see Figure 4-6). The results window shows the integrated Signal-to-Noise Ratio (SNR) as a function of the receive channel number. The PASS/FAIL status should show PASS if all coil elements are functioning properly.

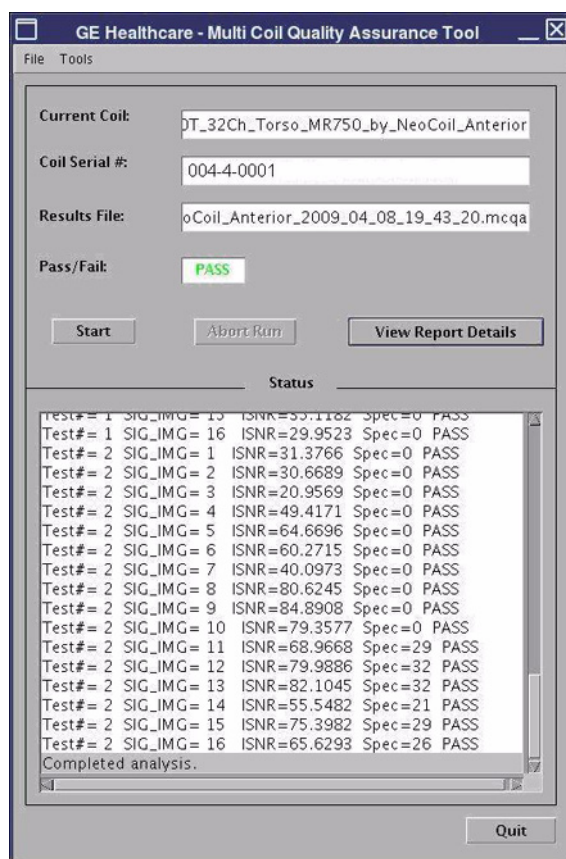


Figure 4-6 MCQA tool displaying the test result for the anterior coil

7. Upon startup of the MCQA tool, or during execution of the test procedure, you may see a phantom placement and landmarking message (see Figure 4-4). Click on **Yes** to continue the test, or **No** to cancel the test so that you can make any adjustments or corrections that may be required.

This page intentionally left blank.

Chapter 5

Coil maintenance

Introduction

To maintain the highest possible image quality, and to protect your patients and yourself from potential injury or illness, it is important to keep the coil clean and well maintained. In this chapter, we discuss basic methods for keeping the coil safe and sanitary. We also provide you with a list of accessories that will enhance the utility of your coil, and you'll find contact information in this section should you have questions or comments you would like to pass along.

Coil care

**WARNING:**

Detach coil connector from the scanner before attempting to clean the coil. Do not touch connectors with bare fingers. Never press sharp objects against connector surface. Do not reattach connector after cleaning until the coil has dried completely. Electric shock may occur if the coil is attached to the system during cleaning or when it is still wet.

**CAUTION:**

Do not spray or pour cleaning solution directly onto the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that may be damaged by the solution. Under no circumstances should the coil be placed into any type of sterilizer. It should be cleaned only according to the procedure described below.

The coil and patient comfort pads must be cleaned and stored using following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- Should your coil need to be returned to us for service, wipe it down with a 10% bleach solution (as described above) to minimize risk of exposure to potentially infectious agents.
- Store the coil in an air conditioned scan or equipment room that meets the temperature and humidity parameters listed in Table 1-2 of this manual.

Store the coil flat. Do not store on edge or on end. To store the coil, a storage space that exceeds the dimensions is required:

- Anterior: 36.2" L x 21.1" W x 1.5" H (91.9 cm x 53.6 cm x 3.8 cm)
- Posterior: 32.5" L x 17.3" W x 2.5" H (82.5 cm x 43.9 cm x 6.4 cm)

Accessories

(can be ordered from NeoCoil)

Part Name	Part Number
Strap assembly	NC004034
Instructions for Use	NC004IFU
Posterior patient pad	NC004033
Chin pad	NC004006
Anterior patient pad	NC004007
Cable pad	NC004008
Pad kit (includes all 4 pads)	NC004012

Contact information

GE Healthcare Americas	
USA	800-558-5102
Canada	800-668-0732
GE Healthcare Europe	
	(33) 1-41-19-76-76
GE Healthcare Asia/Australia	
China	86-21-62192228
Taiwan	886-2-2505-7900
Singapore	65-291-8528
Australia	61-2-9975-5501
Japan	81-120-48-2630
Korea	82-31-740-6119
India	91-80-845-2923
GE Healthcare Web Site	
	www.gehealthcare.com
NeoCoil Corporation	
	1-888-674-2645
NeoCoil Website	
	www.neocoil.com
Manufacturer's name and address	
	NeoCoil N27 W23910-A Paul Rd. Pewaukee, WI 53072

This page intentionally left blank.